2. Encapsulation, Headers, and MTU

02. Encapsulation, Headers, and MTU

Phase: Phase 1 — Models and packets
Estimated time: 1–2 days
Slug: `02-encapsulation`

Learning objectives

- Explain encapsulation/decapsulation
- Define MTU and fragmentation issues
- Use ping to probe path MTU ideas
- Recognize common header sizes intuitively

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills. Type every command yourself and keep a lab notes file.

Why this matters

After enabling a VPN, large uploads hang while small pings work. That is classic MTU/encapsulation pain: each tunnel header steals bytes from the payload budget. Understanding encapsulation and path MTU keeps you from blaming the application for silent blackholes.

Core concepts

Encapsulation wraps upper-layer payloads:

```
[Eth][IP][TCP][HTTP...]
```

- **MTU**: max payload on a link (Ethernet often **1500**).
- VPN/overlay headers reduce effective MTU (e.g. 1420).
- DF (don't fragment) + blocked ICMP "Packet Too Big" → blackhole.
- Rough sizes: Ethernet header 14 (+4 VLAN), IPv4 ~20, TCP ~20 — intuition for MSS clamping.

Worked commands

Run these on a lab host. Linux VM preferred for `ip`/`ss`/`tcpdump`/`nft`; macOS is fine for many `curl`/`dig`/`ping` drills. Your addresses and timings will differ — match the *shape* of the output and the conclusion you draw.

Example 1

```
ip link show | sed -n '1,12p'
# look for mtu 1500 / 9000 / 1420
```

Sample output:

```
2: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 ...
3: wg0: ... mtu 1420 ...
```

Interpretation: Interface MTU is local. Path MTU can be lower than any single hop you admin.

Example 2

```
ping -c 2 -M do -s 1472 1.1.1.1 2>&1 | head -8 || ping -c 2 -D -s 1472 1.1.1.1
2>&1 | head -8
```

Sample output:

```
1472 + 28 = 1500 on IPv4; success means path accepted DF full-size frames (or
middlebox lies).
```

Interpretation: 1472 data bytes + 20 IPv4 + 8 ICMP ≈ 1500. Failure after VPN strongly suggests overhead.

Example 3

```
sudo tcpdump -ni any -c 5 icmp
```

Sample output:

```
IP a > b: ICMP echo request
IP b > a: ICMP echo reply
```

Interpretation: Confirm ICMP is not blocked if you rely on PMTUD. Many ‘mystery hangs’ are PTB filtered.

Mini-lab

1. Record MTU on eth0 and any tunnel interface.
2. Run DF ping at 1472 and at 1400; note difference with VPN on/off.
3. Capture 5 ICMP packets while pinging; identify request/reply.

Failure modes

Symptom	Likely cause	Next check
Large transfers hang, small OK	MTU/PMTUD blackhole	DF ping + lower MSS
VPN works for SSH not HTTPS uploads	Effective MTU	Check tunnel MTU
No ICMP PTB	Firewall too strict	Allow PTB or clamp MSS
Capturing wrong iface	Missed packets	tcpdump -ni any briefly

Study tip for Encapsulation, Headers, and MTU

Before each new command, write a one-line hypothesis. If the result falsifies it, update the hypothesis instead of stacking random tools. Note equivalents across OS (`ip` ↔ `ifconfig`/`route`, `ss` ↔ `netstat`). Re-run two commands from this lesson tomorrow from memory and explain the sample outputs out loud.

Evidence habit

When pages are down, prefer evidence (`ip route get`, resolver answer, packet capture, listener list) over intuition. Document **command → output → conclusion** in a lab notebook so the same path is reusable on-call. Change one variable at a time; write a one-line hypothesis before each new tool.

Common mistakes

- Ignoring MTU after adding VPN/overlays
- Blackhole path MTU (DF set, ICMP blocked)
- Capturing on the wrong interface

Exercises

Work in `exercises/02-encapsulation/`. Read that folder's `README.md` before coding.

```
cd exercises/02-encapsulation
chmod +x test.sh exercise.sh 2>/dev/null || true
./test.sh
```

What you will build

Implement `ethernet_typical_mtu`, `vpn_overhead_effect` (returns `lowers-MTU`), and `df_ping_purpose` (`probe-path-MTU`).

Study loop

- **A (Drill):** make tests pass
- **B (Apply):** run 2–3 real commands from this lesson in your lab and note outputs
- **C (Challenge):** rewrite answers from memory next day

Do not open `solutions/02-encapsulation/` until after an honest attempt.

Self-check

Answer without scrolling up:

1. What does MTU mean?
2. Why do VPNs lower effective MTU?
3. Name a capture tool

Further reading

- https://www.rfc-editor.org/rfc/rfc1191